

Domination and Independent Domination on Probe Interval Graphs

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Abstract

A probe interval graph G is a graph of which the vertices are partitioned into a set $\mathbb{P}$ of *probes* and an independent set $\mathbb{N}$ of *nonprobes* such that G can be embedded into an interval graph by adding some edges between vertices of $\mathbb{N}$. In this paper we propose polynomial-time algorithms using dynamic programming for solving the DOMINATION and the INDEPENDENT DOMINATION problems on probe interval graphs.

1 Introduction

Probe interval graphs, PIGs, were introduced in [17, 18, 19] to model certain problems in physical mapping of DNA. The application in molecular biology is the problem of reconstructing the arrangement of fragments of DNA taken from multiple copies of the same genome. The results of laboratory tests tell us which pairs of fragments occupy intersecting intervals of the genome. The genetic information is physically organized in a linear arrangement, and, when full information is available, an arrangement of intervals can be created in linear time [1].

More recently, a variant that makes more efficient use of laboratory resources has been studied. A subset of the fragments is designated as probes, and for each probe one may test all nonprobe fragments for intersection with the probe.

In graph-theoretic terms, the input to the problem is a graph G and a subset of probe vertices. The other vertices, the nonprobes, form an independent set in G . The objective is to add edges between certain nonprobe vertices such that the graph becomes an interval graph. This problem has been solved successfully by Johnson and

Spinrad [15] in $O(n^2)$ time (where n is the number of vertices of the graph). Soon after, using modular decomposition, McConnell and Spinrad showed that, alternatively, this could be solved in $O(n + m \log n)$ time (where m is the number of edges of the graph) [16]. The problem of recognizing PIGs when the partition of the vertex set is *not* part of the input was recently solved in [3].

In this paper we give the first polynomial-time algorithms for solving the DOMINATION and INDEPENDENT DOMINATION problems for the class of probe interval graphs.

2 Preliminaries

Let $G = (V, E)$ be a simple and undirected graph where V and E are the vertex and edge sets of G , respectively. We write $n = |V|$ for the number of vertices and $m = |E|$ for the number of edges. For our convenience we denote an edge as (x, y) rather than $\{x, y\}$ and say that x and y are *adjacent* in G . If x and y are adjacent we say that they are the *endvertices* of the edge. We only deal with graphs that are *finite*, i.e., the set of vertices is finite. The complement of a graph G , denoted as $\overline{G}$, is the graph in which two vertices are adjacent exactly when they are not adjacent in G .

For a vertex x we write $N(x) = \{y \in V \mid (x, y) \in E\}$ for the set of *neighbors* of x in G , and for a subset $W \subseteq V$ we write $N(W) = \cup_{x \in W} N(x)$. For a vertex x we write $N[x] = N(x) + x$ for its *closed neighborhood*.

For two sets A and B of elements of a common universe, we write $A + B$ and $A - B$ instead of $A \cup B$ and $A \setminus B$ respectively. For a set A and an element x we write $A - x$ instead of $A \setminus \{x\}$ and $A + x$ instead of $A \cup \{x\}$.

For a graph $G = (V, E)$ and a subset $S \subseteq V$ of vertices, we write $G[S]$ for the subgraph of G *induced* by S . For a subset $W \subseteq V$ of vertices of a graph

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$G = (V, E)$ we write $G - W$ for the graph $G[V - W]$, i.e., the subgraph induced by $V - W$. For a vertex x we write $G - x$ rather than $G - \{x\}$. For other conventions on graph-related notation we refer to any standard textbook. For graph classes not defined here we refer to [2, 6, 14].¹

Definiton 1. A graph $G = (V, E)$ is an interval graph if there exists a set $F = \{I_v \mid v \in V\}$ of intervals of the real line such that $I_u \cap I_v \neq \emptyset$ iff $uv \in E$. F is also called the interval model of G .

Both I_v and v can be used to represent the vertex v in an interval graph. Let $l(v)$ and $r(v)$ denote the values of the left and right end points of the interval $I_v = [l_v, r_v]$, respectively.

In this paper we consider probe interval graphs:

Definiton 2. A graph $G = (\mathbb{P} + \mathbb{N}, E)$ is a probe interval graph if there exists a set $F' = F_{\mathbb{P}} + F_{\mathbb{N}} = \{I_v \mid v \in \mathbb{P} + \mathbb{N}\}$ of intervals of the real line such that $I_u \cap I_v \neq \emptyset$ iff $uv \in E$ or $uv \notin E$ but $I_u, I_v \in F_{\mathbb{N}}$. F' is the probe interval model of G .

Similarly, I_v and v can be used to represent vertices in a probe interval graph. Note that a probe interval model can be obtained in polynomial time [16].

Probe interval graphs are weakly chordal.² Consequently, they are perfect [5, 11]. For the various consequences that follow when a graph is perfect and a nice appetizer on perfect graphs we refer to [8]. We now turn to the objective of this paper.

Definiton 3. Let $G = (V, E)$ be a graph. A subset $D \subseteq V$ of vertices is called a dominating set if $\forall x \in V \setminus D, x \cap D \neq \emptyset$, i.e., if every vertex is either in D or has at least one neighbor in D .

The DOMINATION problem is that of finding a dominating set of minimum cardinality. For graphs in general this problem is NP-complete, however, for many special graph classes the situation is much better. For interval graphs there exists a linear time algorithm to solve the DOMINATION problem [4]. There are many papers concerned with finding minimum dominating sets in special graphs. For a bibliography on domination we refer to [10, 13].

Often there are additional constraints on the dominating set. For example, it could be required

that the dominating set is also an independent set. Finding the smallest dominating set satisfying this additional constraint is called the INDEPENDENT DOMINATION problem.

3 Efficient algorithms for probe interval graphs

In this section we describe an algorithm that finds a smallest dominating set in a PIG. We assume the input is a probe interval model of G which had been sorted by their right endpoint and all the intervals are labeled from 1 to n by their ordering. For simplicity, we add a dummy interval 0 in G .

To ease the description of our algorithm we introduce some notation. For interval i , let l_i denote its left endpoint and r_i its right endpoint. The set of intervals that have their right endpoint smaller than l_i is denoted as IFB(i), intervals finish before i . Let $a_i \in \text{IFB}(i)$ be the interval that has the largest right endpoint. Those intervals whose right endpoints are larger than l_i but smaller than r_i are elements in IFI(i), intervals finish in i . Let $b_i \in \text{IFI}(i)$ be the interval that has the smallest right endpoint. Let $\text{PD}_1(i)$ denote a partial dominating set of $\{1, \dots, i\}$. Likewise, $\text{PD}_2(i)$ is a partial dominating set of $\{1, \dots, i\}$ that does not contain i . If there is no dominating set without i , we let $\text{PD}_2(i)$ be the universal set. The minimum partial domination set of probe interval graph is $\text{MDS}(n) = \min(\text{PD}_1(n), \text{PD}_2(n))$.

Assume that $j < i$. Let $\text{NI}_s^+(i, j)$, $s = 1$ or 2 , be the set of nonprobe intervals in $\text{IFI}(i)$ but do not be dominated by $\text{PD}_s(j)$. More precisely, if j' is the largest probe interval in $\text{PD}_s(j)$, then $\text{NI}_s^+(i, j) = \{k \mid k \in \mathbb{N} \text{ and } r_{j'} < l_k \text{ and } r_j < r_k < r_i\}$. Similarly, we define $\text{NI}_s^-(i, j) = \{k \mid k \in \mathbb{N}, r_{j'} < l_k, l_j < r_k < r_j < r_i, \text{ and } l_i < r_k\}$.

We use a linked list to store each $\text{PD}_s(i)$ where $s \in \{1, 2\}$ and $i \in \{1, \dots, n\}$ if i is a nonprobe interval. The reason why we do not choose a minimum one to be $\text{PD}_s(i)$ is that those nonprobe intervals added from $\text{NI}_s^+(i, j)$ may be removed later when we meet a probe interval i' . We cannot guarantee which nonprobe intervals would be removed later. Thus, we construct a linked list to store all possible $\text{PD}_s(i)$ for different j . So that we may find an optimal solution for i' by choosing one of these possible $\text{PD}_s(i)$.

¹See also: <http://www.teo.informatik.uni-rostock.de/isgci/> and <http://www.informatik.hu-berlin.de/hougardy/paper/classes.html>.

²A graph is weakly chordal if it does not contain an induced cycle of length more than 4 or the complement of such a cycle [7, 12, 17].

3.1 Domination

The DOMINATION problem, in a probe interval graph G is to find a minimum set S of intervals such that $S + N(S) = V$.

Lemma 3.1. If interval $i \in \mathbb{P}$, then
 $PD_1(i) = i + \min\{PD_1(j) - NI_1^-(i, j), PD_2(j) - NI_2^-(i, j) \mid r_{a_i} \leq r_j < r_i\}$

Proof. Suppose each $PD_s(j)$, $s \in \{1, 2\}$ and $j \in \{0, 1, \dots, i-1\}$ is correct. If there exists an interval k which cannot be dominated by $PD_s(j)$, then $l_k > r_j$. By definition, $r_{a_i} \leq r_j$. Thus k must be dominated by i . On the contrast, if k is not dominated by i , then $r_k < l_i$. Since $r_{a_i} \leq r_j < r_i$ and r_{a_i} is the rightmost endpoint before l_i , $r_k \leq r_{a_i}$. That is, k must be dominated by $PD_s(j)$. Finally, since $NI_s^-(i, j)$, where $s=1$ or 2 , is dominated by i , it can be removed from $PD_1(i)$. Therefore our lemma holds. $\square$

Lemma 3.2. If interval $i \in \mathbb{P}$, then
 $PD_2(i) = \min\{PD_1(j) \mid \text{there is no interval } k \text{ with } r_{b_i} \leq r_j < l_k < r_k < r_i \text{ and } NI_1^+(i, j) = \emptyset\}$

Proof. Similar to Lemma 3.1. $\square$

Lemma 3.3. If interval $i \in \mathbb{N}$, then
 $PD_1(i) = i + \{PD_1(j) + NI_1^+(i, j), PD_2(j) + NI_2^+(i, j) \mid r_{a_i} \leq r_j < r_i\}$

Proof. Suppose each $PD_s(j)$, $s \in \{1, 2\}$ and $j \in \{0, 1, \dots, i-1\}$ is correct. It is not to see that if k is nonprobe that intersects with i and is not in $NI_1^+(i, j)$, then k is dominated by $\{PD_1(j)\}$. Since $NI_1^+(k+1, j)$ can be removed when we meet a probe interval. Thus we keep all possible $PD_1(i)$'s (in this case, local optimal is not global optimal). The remaining is similar to Lemma 3.1. $\square$

Lemma 3.4. If interval $i \in \mathbb{N}$, then
 $PD_2(i) = \{PD_1(j) + NI_1^+(i, j) \mid j \in P \text{ and there is no probe interval } k \text{ with } r_{b_i} \leq r_j < l_k < r_k < r_i\}$

Proof. Similar to Lemma 3.3. $\square$

Algorithm: Domination

Input: A probe interval model F of G .

Output: A minimum dominating set, MDS.

- 1: Initially, for each $i \in \{1, \dots, n\}$, we compute a_i and b_i , and set $PD_1(i) = PD_2(i) = F$.
- 2: $PD_1(0) = \{0\}$; $PD_2(0) = F$;

- 3: **for** $i = 1, \dots, n$ **do**
- 4: **if** $i \in \mathbb{P}$ **then**
- 5: **for** each j with $r_{a_i} \leq r_j < r_i$ **do**
- 6: compute $NI_1^-(i, j)$ and $NI_2^-(i, j)$;
- 7: $PD_1(i) = \min\{PD_1(i), i + PD_1(j) - NI_1^-(i, j), i + PD_2(j) - NI_2^-(i, j)\}$;
- 8: **for** each j with no interval k s.t. $r_{b_i} \leq r_j < l_k < r_k < r_i$ and $NI_1^+(i, j) = \emptyset$ **do**
- 9: $PD_2(i) = \min\{PD_2(i), PD_1(j)\}$;
- 10: **else**
- 11: **for** each j with $r_{a_i} \leq r_j < r_i$ **do**
- 12: compute $NI_1^+(i, j)$ and $NI_2^+(i, j)$;
- 13: store $i + PD_s(j) + NI_s^+(i, j)$ to the linked list associated with $PD_1(i)$;
- 14: **for** each $j \in P$ with no probe interval k s.t. $r_{b_i} \leq r_j < l_k < r_k < r_i$ **do**
- 15: compute $NI_1^+(i, j)$;
- 16: store $PD_1(j) + NI_1^+(i, j)$ to the linked list associated with $PD_2(i)$;
- 17: $MDS = \min\{PD_1(n), PD_2(n)\}$

By Lemmas 3.1 to 3.4, we have the following theorem.

Theorem 1. The DOMINATION problem for a probe interval graph can be solved in polynomial time.

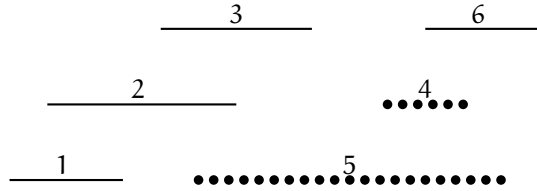


Figure 1: A probe interval graph G with nonprobe intervals represented as dotted lines and probe intervals represented as straight line.

We give an example shown in Figure 1 for describing our algorithm. In figure 1, intervals are labeled according to their right endpoints. Initially, $PD_1(0) = \{0\}$ and $PD_2(0) = F$. Interval 1 is a probe interval, by our algorithm, $PD_1(1) = \{0, 1\}$ and $PD_2(1) = F$. When the second interval 2 which is a probe interval comes in, $PD_1(2)$ would be $\{0, 2\}$, and $PD_2(2) = \{0, 1\}$. For interval 3, $PD_1(3) = \{0, 1, 3\}$, $PD_2(3) = \{0, 2\}$. Further, for each $i = \{1, 2, 3\}$, $NI_s^+(i, j) = \emptyset$ with $s = 1$ or 2 .

Now we consider interval 4 which is a nonprobe interval. By our algorithm, $PD_1(4) = \{0, 2, 4\}$, $PD_2(4) = F$ while $NI_1^+(4, 3) = \emptyset$. For interval 5 which is also a nonprobe interval. we obtain $PD_1(5) = \{5\} + PD_1(3) + NI_1^+(5, 3) = \{5\} + \{0, 1, 3\} + \{4\}$ or $PD_1(5) = \{5\} + PD_2(3) + NI_2^+(5, 3) = \{5\} + \{0, 2\} + \{4\}$ and $PD_2(5) =$

$\min\{PD_1(2) + NI_1^+(5, 2), PD_1(3) + NI_1^+(5, 3)\} = \min\{\{0, 2\} + \{4\}, \{0, 1, 3\} + \{5\}\} = \{0, 2, 4\}$.

Finally, for interval 6 which is a probe interval, we obtain that $PD_1(6) = \min\{\{6\} + PD_1(3) + NI_1^+(6, 3), \{6\} + PD_2(3) + NI_2^+(6, 3)\} = \min\{\{6\} + \{0, 2\}, \{6\} + \{0, 1\} + \{4\} - \emptyset, \{6\} + \{0, 2\} - \emptyset\} = \{0, 2, 6\}$ and $PD_2(6) = PD_1(5) = \{0, 2, 4, 5\}$. Finally, we obtain the solution $MDS = \min\{PD_1(6), PD_2(6)\} = \min\{\{0, 2, 6\}, \{0, 2, 4, 5\}\} = \{0, 2, 6\}$.

3.2 Independent domination

The INDEPENDENT DOMINATION SET problem asks for a domination set S of minimum cardinality such that S is an independent set. The following lemmas for independent domination are similar to those for domination. We omit the proof in this extended abstract.

Lemma 3.5. If interval $i \in \mathbb{P}$, then $IPD_1(i) = i + \min\{IPD_1(j) - NI_1^-(i, j), IPD_2(j) - NI_2^-(i, j) \mid j = a_i\}$.

Lemma 3.6. If interval $i \in \mathbb{P}$, then $IPD_2(i) = \min\{IPD_1(j) \mid \text{there is no interval } k \text{ with } r_{b_i} \leq r_j < l_k < r_k < r_i \text{ and } NI_1^+(i, j) = \emptyset\}$.

Lemma 3.7. If interval $i \in \mathbb{N}$, then $IPD_1(i) = i + \{IPD_s(j) + NI_s^+(i, j) \mid j = a_i \text{ and } s \in \{1, 2\}\}$.

Lemma 3.8. If interval $i \in \mathbb{N}$, then $IPD_2(i) = \{IPD_1(j) + NI_1^+(i, j) \mid j \in P \text{ and there is no probe interval } k \text{ with } r_{b_i} \leq r_j < l_k < r_k < r_i\}$.

The following is our algorithm.

Algorithm: Independent-domination

Input: A probe interval model F of G .

Output: A minimum independent dominating set, MIDS.

- 1: Initially, for each $i \in \{1, \dots, n\}$, we compute a_i and b_i , and set $IPD_1(i) = IPD_2(i) = F$.
- 2: $IPD_1(0) = \{0\}$; $IPD_2(0) = F$;
- 3: **for** $i = 1, \dots, n$ **do**
- 4: **if** $i \in \mathbb{P}$ **then**
- 5: compute $NI_1^-(i, j)$ and $NI_2^-(i, j)$;
- 6: $IPD_1(i) = \min\{IPD_1(i), i + IPD_1(a_i) - NI_1^-(i, a_i), i + IPD_2(a_i) - NI_2^-(i, a_i)\}$;
- 7: **for** each j with no interval k s.t. $r_{b_i} \leq r_j < l_k < r_k < r_i$ and $NI_1^+(i, j) = \emptyset$ **do**
- 8: $IPD_2(i) = \min\{IPD_2(i), IPD_1(j)\}$;
- 9: **else**
- 10: compute $NI_1^+(i, a_i)$ and $NI_2^+(i, a_i)$;

- 11: $IPD_1(i) = \min\{IPD_1(i), i + IPD_1(a_i) + NI_1^+(i, a_i), i + IPD_2(a_i) + NI_2^+(i, a_i)\}$;
- 12: **for** each $j \in P$ with no probe interval k s.t. $r_{b_i} \leq r_j < l_k < r_k < r_i$ **do**
- 13: store $IPD_1(j) + NI_1^+(i, j)$ to the linked list associated with $IPD_2(i)$.
- 14: MIDS = $\min\{IPD_1(n), IPD_2(n)\}$

Theorem 2. The INDEPENDENT DOMINATION problem for a probe interval graph can be solved in polynomial time.

4 Concluding remarks

In this paper we presented the first polynomial time algorithms for solving some domination problems on probe interval graphs. Since this is a new class of perfect graphs, many problems remain open. Since the class is perfect, the 4 “basic” NP-complete problems CLIQUE, INDEPENDENT SET, CLIQUE COVER, and CHROMATIC NUMBER are all solvable in polynomial time [9]. However, the general method for these problems uses the ellipsoid method which is impractical.

One open problem that we like to mention is the *coloring* problem. Since the chromatic number equals its clique number, this is solvable in linear time for probe interval graphs (when the partition is part of the input). However, finding an actual optimal coloring is perhaps a more difficult problem.

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